

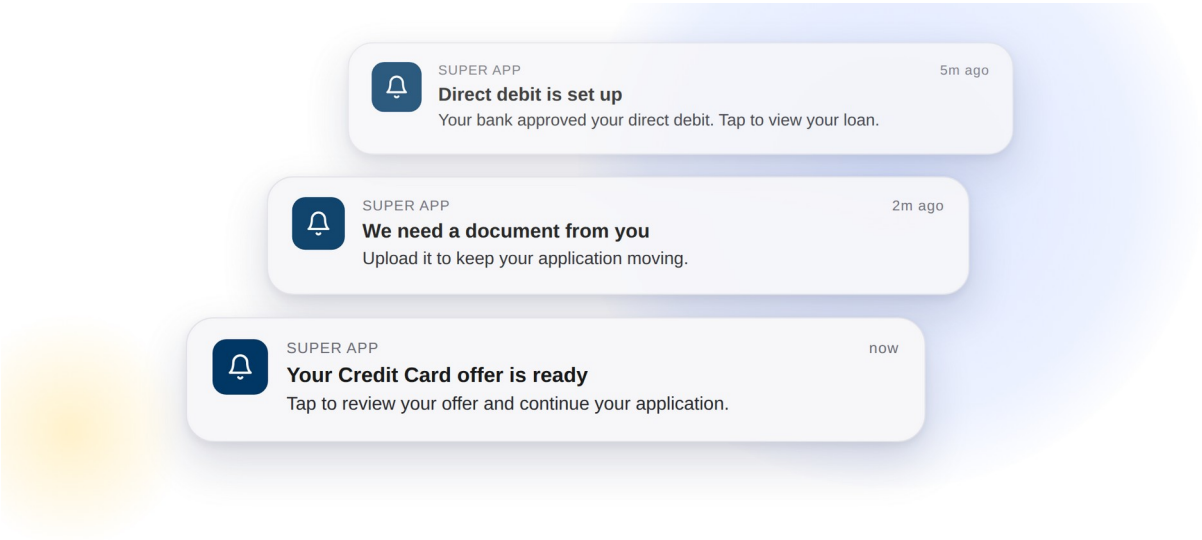


BUSINESS REQUIREMENTS DOCUMENT

Push Notification

Customer journey alerts in the Super App — Credit Card, Personal Loan and CASA

V1.2 / 06 October 2026



VERSION HISTORY

Date	Version	Author	Change Description
28-09-2026	1.0	Hailey (Appro)	Created BRD
28-09-2026	1.1	Hailey (Appro)	Trigger events and push content in one table; push results tracked in the backend only, not on the audit trail; sample request and response exactly as the middleware specification
06-10-2026	1.2	Hailey (Appro)	Business review of 29/09 incorporated: updated wording for P01, P04, P05 and P09; tap behaviour and destination screens; device and session scenarios; reminder stop rules and sending time; retry baseline and error codes per middleware specification v0.2 (05/10); duplicate-prevention rule BR5; acceptance criteria; future enhancements (post-MVP, dedicated development team)

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1. FEATURE OVERVIEW AND KEY COVERED AREAS

This document describes the requirements for **Push Notification** in the Reem Bank customer journey for **Credit Card, Personal Loan and CASA**, on iOS and Android.

Today the customer learns about decisions and required actions on an application — a credit decision, a Direct Debit Authority response, the end of the cooling-off period — by **email and in-app notification only**. When the customer is outside the Super App, nothing reaches the device, and the application waits until the customer returns.

Push Notification adds a device-level alert at each agreed trigger event. The Appro platform sends the push request to the Reem Bank middleware Push API, following the same integration pattern as the existing SMS and email integration. When the customer taps the notification, the Appro journey opens at the screen where the application continues.

This BRD covers the following areas:

- **Trigger events and push content** — eleven notifications, each with its title, body and the screen it opens
- **Language** — English or Arabic, following the customer’s current language
- **Middleware integration** — the Push API request, headers and response
- **Response handling and automatic retry**
- **Business rules** — personal data and excluded templates

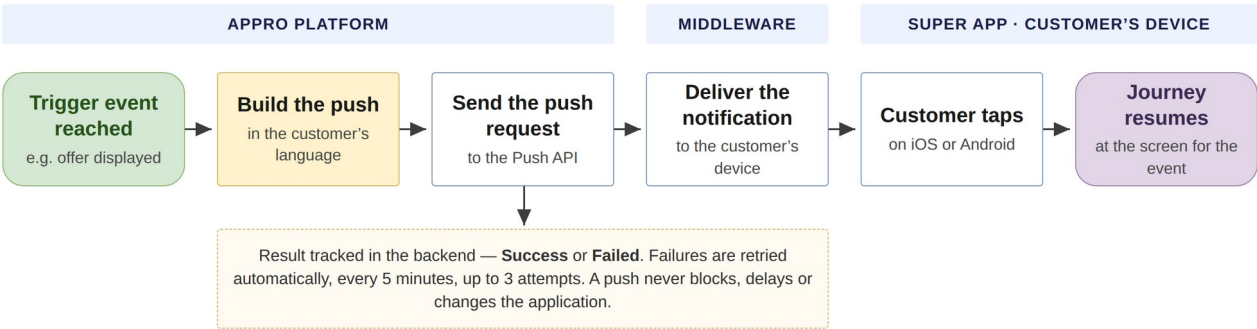


Figure 1 — How a push reaches the customer (green = start, amber = system action, blue = step, purple = end)

Key steps

1. A trigger event is reached on the application, for example an offer is displayed or a document is requested.
2. The system builds one push for the event: the Push ID, the title and the body, in the customer’s current language.
3. The system sends the push request to the Reem Bank middleware Push API.
4. The middleware delivers the notification to the customer’s device through the Super App.
5. The customer taps the notification, and the Appro journey opens at the screen for that event.
6. The result is tracked in the backend. A failed push is retried automatically and never blocks the application.

Note: Push content is fixed per notification and is not configurable in Super Portal in this release. Customer actions on a notification (open, dismiss, ignore) are not tracked. Existing email and in-app notifications are unchanged. Business-managed configuration and Super Portal visibility are planned as post-MVP enhancements (section 9).

2. SCOPE

2.1 Product scope

Credit Card, Personal Loan and CASA, on **iOS and Android**. Each notification applies only to the products listed against it in section 3.

2.2 In-scope notifications

- **P01** — Offer ready (Credit Card, Personal Loan)
- **P01 (CASA)** — Account approved in principle (CASA)
- **P02** — Offer expiry reminder, once per day for 15 days (all products)
- **P03** — Offer expired (all products)
- **P04** — Update on the application after a rejection (all products)
- **P05** — Document requested from the customer (all products)
- **P06** — Application active again after the cooling-off period (Credit Card, Personal Loan)
- **P08** — Direct debit set up (Personal Loan)
- **P09** — Direct debit rejected by the customer's bank (Personal Loan)
- **P10** — Loan on its way after disbursement (Personal Loan)
- **P11** — Product ready: card issued (Credit Card, CASA) or loan approved (Personal Loan)

2.3 Not in scope

- OTP messages — never sent as push (BR3).
- Bank and internal-user notifications — never sent as push (BR4).
- Configuring push content in Super Portal — content is fixed per notification in this release.
- Tracking customer actions on a notification — open, dismiss or ignore.
- Changes to the existing email and in-app notifications — they remain unchanged (BR1).

3. TRIGGER POINTS AND PUSH CONTENT

3.1 Trigger events and push content

A push is fired at the same trigger point as the existing email and in-app notification for the same event. Existing email and in-app notifications remain unchanged.

Push ID	Priority	Applies to	Trigger event	Title (EN, max 40 characters)	Body (EN, max 100 characters)
P01	P0	CC · PL	AIP given (at least one offer displayed), or Credit Queue approves / overrides the application	Your offer is ready	Review your offer and continue your application.
P01 (CASA)	P0	CASA	Account application approved in principle	Your account is approved	Tap to continue and complete your account opening.
P02	P0	CC · PL · CASA	Scheduled reminder once per day for 15 days, starting after the first offer is displayed	Your offer expires in %%COUNTING_DOWN%% days	Complete your application before your approval expires.
P05	P0	CC · PL · CASA	Credit or Sales team returns the case requesting documents from the customer	One more document is needed	Tap to see which document is needed and how to share it. Wording to be confirmed with Business once the document channel is agreed.
P06	P0	CC · PL	Cooling-off end date reached (start + 5 days, configurable) with no customer action; application returns to Approved In Principle	Your application is active again	Your cooling-off period has ended. Tap to continue or cancel.
P09	P0	PL	Customer's bank rejects the Direct Debit Authority	Your direct debit needs attention	Your bank could not approve the request. Tap to review the next steps.
P04	P1	CC · PL · CASA	Queue user rejects (Credit, Sale, Risk or Compliance), Rule Engine filtration fails, or minimum income not met	There's an update on your application	Tap to view the latest status of your application.
P08	P1	PL	Customer's bank accepts the Direct Debit Authority	Direct debit is set up	Your bank approved your direct debit. Tap to view your loan.
P10	P1	PL	Disbursement Checker approves and releases the disbursement	Your loan is on its way	Funds will reach your account shortly. Tap for details.
P11	P1	CC · PL · CASA	Card issued (CC, CASA) or loan approved (PL)	Your %%PRODUCT_TYPE%% is ready	Tap to see what happens next.
P03	P2	CC · PL · CASA	Offer validity (30 days) elapses with no acceptance	Your offer has expired	You can reapply whenever you're ready.

3.2 Content rules

- [%%PRODUCT_TYPE%%] and [%%COUNTING_DOWN%%] must be replaced with actual values before sending. No placeholder is ever shown to the customer.
- The title must not exceed **40 characters** and the body **100 characters**, after replacement.
- Content is fixed per Push ID and is not configurable in Super Portal in this release.

3.3 Language

- The push is sent in **English or Arabic**, following the language the customer is currently using.
- Arabic title and body for every notification – **To be provided by RB Business**

3.4 Tap behaviour and destination screens

Every push carries an action type (section 4.3). APPRO_JOURNEY opens the Appro journey at the destination screen below. NONE — used for P03 and P09, per the alignment call of 05/10 — opens the Super App without forcing a screen. If the customer is logged out, the Super App completes the login first, then opens the destination screen.

Push ID	Action type	On tap, opens at	If no longer active
P01	APPRO_JOURNEY	Offer review screen	Application status screen
P01 (CASA)	APPRO_JOURNEY	Account opening continuation screen	Application status screen
P02	APPRO_JOURNEY	Offer review screen	Application status screen
P03	NONE	Super App — no Appro screen is opened	—
P04	APPRO_JOURNEY	Application status screen	Application status screen
P05	APPRO_JOURNEY	Document request screen	Application status screen
P06	APPRO_JOURNEY	Offer review screen	Application status screen
P08	APPRO_JOURNEY	Loan status screen	Application status screen
P09	NONE	Super App — no Appro screen is opened	—
P10	APPRO_JOURNEY	Loan status screen	Application status screen
P11	APPRO_JOURNEY	Application status screen	Application status screen

The Super App hand-over of action.id and action.value to the Appro journey, and keeping the destination through the login — TBC by Avanza.

3.5 Device and session scenarios

Each scenario has one owner. Appro rows are covered by this BRD; the Super App and middleware rows are **TBC by Avanza**.

Scenario	Owner	Behaviour
Customer logged out or session expired	Super App	The Super App completes the login first, then opens the destination screen of section 3.4.
Customer changed device, or multiple registered devices	Super App	The push follows the device-token registration held by the Super App.
Push permission disabled on the device	Super App	No device alert. The application and the existing email and in-app notifications continue unchanged.

Scenario	Owner	Behaviour
Device token invalid or expired	Middleware	The middleware returns 404 UNREGISTERED — tracked in the backend as Failed, no retry (section 4.5).
Application completed, cancelled, withdrawn, rejected or expired	Appro	No further push is sent and reminders stop immediately (section 3.6). On a late tap, the journey opens the application status screen (section 3.4).
Customer no longer eligible	Appro	A push is built only at its trigger event on a live application — no event, no push.

3.6 Reminder and stop rules (P02)

- P02 is sent once per day for 15 days, starting after the first offer is displayed. The interval and the maximum count are backend configuration parameters (default: daily, 15).
- P02 stops immediately when the customer selects an offer, or when the application is completed, cancelled or withdrawn, rejected or expired, or the product is booked or issued.
- One push per Push ID per trigger occurrence per application (BR5).

3.7 Sending time

- Event-driven pushes are sent immediately at the trigger event.
- Scheduled pushes — the P02 reminder and P03 (offer expired) — are released at 10:00 UAE time (backend configuration parameter).

4. PUSH NOTIFICATION API INTEGRATION

4.1 Request details

Item	Value
Method	POST
Endpoint	http://<MW URL>/api/notifications/push TBC by Avanza
Authentication	Bearer token TBC by Avanza

4.2 Header parameters

Parameter	Type	Required? (Y/N)	Description	Values / Data Source
stan	String	Y	Unique system audit trace number in each request	Generated by Appro, unique per request. 6–12 characters (specification v0.2).
Authorization	String	Y	Authorization header	Bearer <token>
Content-Type	String	Y	Request format	application/json
channel_id	String	Y	Channel id of the calling system	TBC by Avanza - value assigned to Appro

4.3 Request parameters

Parameter	Type	Required? (Y/N)	Description	Values / Data Source
requestId	String	Y	Unique message id	Generated by Appro, unique per push. Every retry reuses the same requestId. TBC by Avanza: middleware de-duplication on the requestId.
timestamp	String	Y	Request date-time	ISO 8601 UTC with milliseconds, e.g. 2026-09-28T06:49:02.366Z
type	String	Y	Notification type	Fixed: APPRO_NOTIFICATION
action.id	String	Y	Push identifier	Push ID from section 3, e.g. P01. P01 (CASA) is sent as P01 with action.value = CASA
action.type	String	Y	NONE / POPUP / APPRO_JOURNEY	APPRO_JOURNEY: a tap opens the Appro journey (section 3.4). NONE: no forced action — the tap opens the Super App; used for P03 and P09 (call of 05/10). POPUP: not used by Appro. TBC by Avanza: Super App hand-over of action.id and action.value to the Appro journey.
action.value	String	Y	Product	CREDIT_CARD / PERSONAL_LOAN / CASA, from the application.
recipient.mobilePhone	String	Y	Customer mobile number	From the application. Format 9715XXXXXXX — country code 971, no plus sign (specification v0.2).
recipient.customerId	String	Conditional	Customer CIF — required if ETB	ETB: customer CIF. NTB: not sent.
recipient.email	String	N	Customer email	No mapping – Optional field
content.title	String	Y	Push title	Per section 3.1, in the customer's current language
content.body	String	Y	Push body	Per section 3.1, in the customer's current language
content.message	String	N	Notification message	Sent empty (""). Pop-up content, used only with action type POPUP (specification v0.2).
content.language	String	Y	Language — "EN" / "AR"	Customer's current language — "EN" or "AR" (specification v0.2).

4.4 Response parameters

Parameter	Type	Description	Values / Handling
messageId	String	Message id returned by the middleware	Tracked in the backend

Parameter	Type	Description	Values / Handling
error.code	String	Result code	"000" = Processed Ok. Full error-code list in section 4.5.
error.message	String	Result description	Tracked in the backend with error.code

4.5 Error codes (middleware specification v0.2)

Code	Description	Appro handling
000	PROCESSED OK	Push sent — tracked in the backend as Success.
403	SENDER_ID_MISMATCH	Final, no retry — configuration issue, raised with Avanza. Tracked as Failed.
404	UNREGISTERED	Final, no retry — the device token is not registered. Tracked as Failed.
429	QUOTA_EXCEEDED	Retried per section 5.
500	INTERNAL SERVER ERROR	Retried per section 5.

Retry classification proposed by Appro — TBC by Avanza.

4.6 Open items with Avanza

Open as of 06 October 2026 (our questions of 28/09; specification v0.2 and the alignment call of 05/10 closed the rest):

- Middleware endpoint and credentials per environment. TBC by Avanza
- channel_id value assigned to Appro. TBC by Avanza
- De-duplication on the requestId when a retry resends the same id. TBC by Avanza
- Super App hand-over of action.id and action.value to the Appro journey, and keeping the destination through the login. TBC by Avanza

5. RESPONSE HANDLING AND RETRY

- If the HTTP code is **20x** and [error.code] = **"000"** (Processed Ok) → the push is sent → tracked in the backend as **Success** → the journey proceeds.
- Else → the system retries automatically after **5 minutes**, up to **3 attempts**. The interval and the number of attempts are backend configuration parameters.
 - "Else" covers a timeout, any non-20x HTTP code and any [error.code] other than "000".
 - Error codes 429 and 500, and a timeout, are retried. 403 and 404 are final and are not retried (section 4.5).
 - Every retry reuses the same requestId, so a retry never creates a duplicate notification (BR5).
- If the push is still not successful after the last attempt → tracked in the backend as **Failed** → the journey proceeds. A push failure must not block, delay or change the application status.
- The system tracks the API delivery result only, in the backend; it is not shown in the Super Portal audit trail. Customer actions on the notification (open, dismiss, ignore) are not tracked. The backend record per push: Push ID, date and time, number of attempts, final status (Success / Failed) and the failure reason. Until a Super Portal view is available (section 9), Appro support provides the status of a push for a given application on request from the Contact Centre.

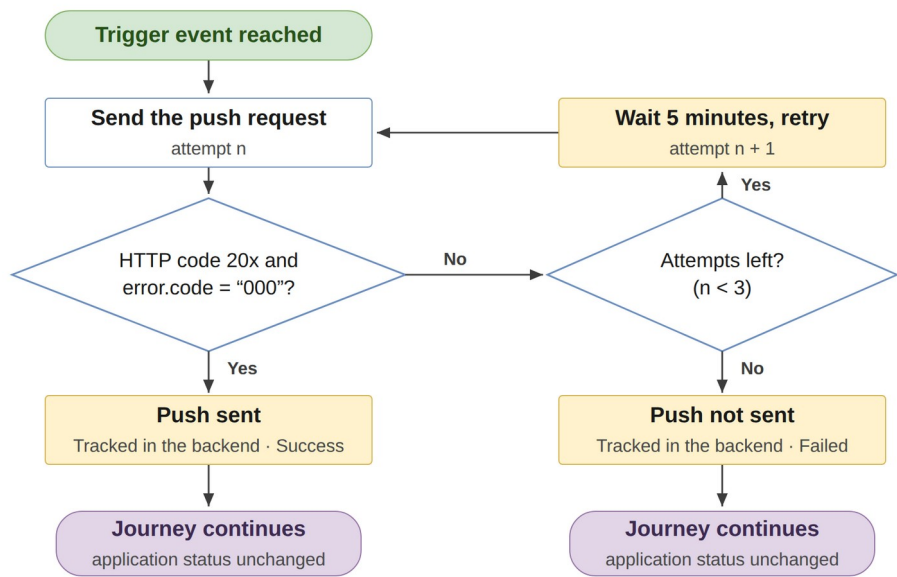
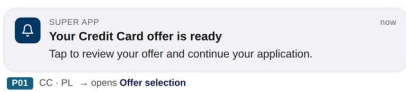
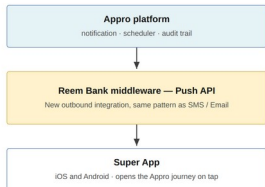
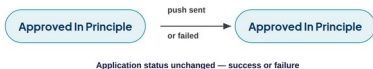


Figure 2 — Response handling and automatic retry (default: retry every 5 minutes, up to 3 attempts)

6. BUSINESS RULES

No.	Rule
BR1	Push is additional; existing email and in-app notifications are unchanged.
BR2	No personal data in the title or body: no name, Application ID, IBAN or amounts.
BR3	OTP content (ET17, ET18, ET35, NT04, NT05) is never sent as push.
BR4	Bank and internal-user templates are never sent as push.
BR5	One notification per event: a push is sent once per Push ID per trigger occurrence per application, and every retry reuses the same requestId — a system retry, timeout or reprocessing never creates a duplicate.

7. IMPACT ANALYSIS

Area	Impact	Screen																														
Customer journey (Super App)	A device notification at eleven trigger events. A tap opens the Appro journey at the screen for that event; existing journey screens are reused as the destinations.																															
Middleware integration	New outbound call from the Appro platform to the Reem Bank middleware Push API, following the SMS and email integration pattern. Endpoint, credentials and channel id per environment: TBC by Avanza.																															
Communication Setup	No change. Existing email and in-app templates keep working as today and remain the trigger reference for each push. Push content is fixed per notification in this release and is not added to Communication Setup.	<table><tr><th>No</th><th>Email Template Name</th><th>Status</th><th>Type</th><th>Send To</th></tr><tr><td>1</td><td>If the Rule Engine is "Status" or "Refer" (move to Credit Team)</td><td>☒</td><td>Client</td><td></td></tr><tr><td>2</td><td>Application cancellation request rejected</td><td>☒</td><td>Bank</td><td></td></tr><tr><td>3</td><td>Application cancellation request completed</td><td>☒</td><td>Bank</td><td></td></tr><tr><td>4</td><td>Application is cancelled</td><td>☒</td><td>Client</td><td></td></tr><tr><td>5</td><td>Customer doesn't receive off cooling period</td><td>☒</td><td>Client</td><td></td></tr></table>	No	Email Template Name	Status	Type	Send To	1	If the Rule Engine is "Status" or "Refer" (move to Credit Team)	☒	Client		2	Application cancellation request rejected	☒	Bank		3	Application cancellation request completed	☒	Bank		4	Application is cancelled	☒	Client		5	Customer doesn't receive off cooling period	☒	Client	
No	Email Template Name	Status	Type	Send To																												
1	If the Rule Engine is "Status" or "Refer" (move to Credit Team)	☒	Client																													
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3	Application cancellation request completed	☒	Bank																													
4	Application is cancelled	☒	Client																													
5	Customer doesn't receive off cooling period	☒	Client																													
Audit trail	No change in this release. Push results are tracked in the backend only; Appro support provides the status of a push on request (section 5). A Super Portal view is a post-MVP enhancement (section 9).	No screen change																														
Application status	No change. A push — sent or not — never changes, blocks or delays the application.																															
Reporting	The API delivery result is tracked. Customer actions on a notification (open, dismiss, ignore) are not tracked in this implementation scope	No screen change																														

8. ACCEPTANCE CRITERIA (UAT)

UAT passes when the criteria below hold; the detailed test cases follow in the UAT plan. The second group is executed end to end with Avanza, because the behaviour sits in the Super App or the middleware.

Verified on the Appro platform

- Each push fires at its trigger event only, for the right product, customer and language, with the exact title and body of section 3.1 — and never for a completed, cancelled, rejected or expired application.
- A tap opens the destination screen of section 3.4, on iOS and Android.
- Arabic content renders correctly, and no personal data appears in any title or body (BR2).
- A failed call is retried every 5 minutes up to 3 attempts; 403 and 404 are not retried; the final failure is tracked with its reason (sections 4.5 and 5).
- No duplicate notification for one event — including on retry, timeout or reprocessing (BR5).
- P02 stops immediately on every stop condition of section 3.6; scheduled pushes respect the 10:00 sending time (section 3.7).
- A push failure never blocks, delays or changes the application (section 5).
- The backend record is complete per section 5 for every sent and failed push.

Verified end to end with Avanza

- Delivery on iOS and Android through the Super App.
- Notification permission disabled, invalid or expired token, changed device, multiple devices (section 3.5).
- Logged-out tap: login first, then the destination screen of section 3.4.
- P03 and P09 (action type NONE) open the Super App only.

9. FUTURE ENHANCEMENTS (POST-MVP)

The items below are agreed as enhancements after the MVP 1.1 sign-off, delivered by the dedicated development team based on launch feedback — not as change requests. The MVP design is scalable, so they are added on the same framework, without redesign.

- Business-managed configuration in Super Portal, wherever technically feasible without source-code modification: enable / disable per notification, English and Arabic content, reminder frequency and maximum count, validity period, priority, trigger and product mapping, and new Push IDs.
- Sending notifications from Super Portal.
- Push visibility in Super Portal: an audit-trail entry or a notification report with the sent / failed status per application.
- Configurable sending hours and customer local time.
- Additional products, journeys and notification types on the same framework.

APPENDIX 1: SAMPLE REQUEST/RESPONSE

Sample request

```
{
  "requestId": "1234567891011121",
  "timestamp": "2020-07-20T06:49:02.366Z",
  "type": "APPRO_NOTIFICATION",
  "action": {
    "id": "P01",
    "type": "APPRO_JOURNEY",
    "value": "CASA"
  },
  "recipient": {
    "mobilePhone": "971528045456 ",
    "customerId": "104278",
    "email": "abc@test.com"
  },
  "content": {
    "title": "CASA Journey",
    "body": "Please resume your CASA journey",
    "message": "",
    "language": "EN"
  }
}
```

Sample response

```
{
  "messageId": "0:1790341509514173%d5881a7ad5881a7a",
  "error": {
    "code": "000",
    "message": "Processed 0k"
  }
}
```

Product mapping

Product on the application	action.value
Credit Card	CREDIT_CARD
Personal Loan	PERSONAL_LOAN
CASA	CASA